Practice Guide Overview

This page brings together the practical lab files for the course. Instead of linking the main course notebooks, it points to the exercise and practice guides in the practice folder.

01 - Spam / Adversarial Text

These practice files build a spam filter from the ground up and then stress-test it with adversarial examples.

- spam/guide_02_01.ipynb — Build a spam filter from scratch with Naive Bayes, TF-IDF, and logistic regression.

- spam/guide_02_02.ipynb — Compare word vectors, sentence embeddings, and llama.cpp-based embeddings for the same spam task.

- spam/guide_02_03.ipynb — Find counterexamples, useful adversarial spam, and defenses that keep the filter robust.

02 - Anomaly Detection

These guides focus on building and attacking anomaly detectors for network traffic and temporal patterns.

- anomaly/guide_03_01.ipynb — Classic anomaly detectors on network connections: z-score, Mahalanobis distance, thresholds, and KDD99 evaluation.

- anomaly/guide_03_02.ipynb — Autoencoders, time-series anomalies, and reconstruction-based detection in an adversarial setting.

- anomaly/guide_03_03.ipynb — Test anomaly detectors as an adversary would: evasion, mimicry, poisoning, and robust defenses.

03 - Malware / ML Evaluation

This folder includes the malware-oriented practice material and dataset exploration for feature analysis.

- malware/notebook.ipynb — Malware feature exploration with API call sequences, hybrid datasets, and lightweight preprocessing.

NOT YET PREPARED

Suggested reading order

1. Start with the spam practice guides in spam.
2. Continue with anomaly detection in anomaly.
3. Finish with the malware notebook in malware.

This page gives a quick map of the practice work. It is meant to leave space for notes between each exercise.